Predictive multiplicity in environmental health

1 Data and model

18 satellite and gridded modalities

3,077 US counties

2003 to 2019

no socioeconomic input

$R^2 = 0.666$

out of fold, county-grouped

2 Two descriptions

formaldehyde, time above P75

rank 1

same field, annual mean

rank 196

night cooling, two forms

20.3× apart

3 Multiplicity

SHAP favours a non-central summary

38 of 47

34.9 expected at random

mutual information favours the mean

35 of 47

R^2, central-only vs non-central-only

0.6079 vs 0.5967

4 What survives

duration advantage vs lifetime

$\rho = -0.42$

nocturnal turning points

dew point 5.54 °C

wet-bulb 7.41 °C

shift across the 100th meridian

≤ 1.7% of range

Attribution records how a model digests a field, not how much information the field holds; physical structure survives the multiplicity.